

EXPERIMENT - 1

AIM - Design web pages for your college containing a description of the courses, departments, faculties, library etc. Use href and list tags.

OBJECTIVE:

Design web pages for a college showcasing courses, departments, faculties, and library details using HTML href and list tags. This experiment aims to develop skills in creating structured, navigable, and user-friendly web pages.

INTRODUCTION:

This experiment focuses on creating organized and user-friendly web pages for a college. Students will use **href** for linking pages and **list tags** to structure content, enhancing their HTML skills.

HTML (HyperText Markup Language)

HTML is the standard markup language used to create the structure of web pages. It uses tags to define elements such as headings, paragraphs, images, lists, and hyperlinks. In this experiment, HTML is used to design different sections of a college website such as courses, departments, faculty, and library using tags like `<a>`, `<ul>`, `<li>`, and `id`.

Hyperlink (href Attribute)

The href attribute is used with the `<a>` tag to create hyperlinks. It helps in navigating between different webpages or different sections of the same webpage.

List Tags (`<ul>`, `<ol>`, `<li>`)

List tags are used to display information in an organized form. Unordered and ordered lists contain list items and are used to present courses, faculty details, and resources clearly.

id Attribute

The id attribute is used to uniquely identify an HTML element. It helps in creating internal links that allow direct navigation to specific sections of a webpage.

CSS (Cascading Style Sheets)

CSS is used to control the presentation and appearance of web pages. It allows developers to apply styles such as colors, fonts, layout, and background images to HTML elements. Internal CSS is commonly used to enhance the visual look of the webpage and make it more user-friendly.

Webpage

A webpage is a digital document displayed in a web browser and written using HTML and CSS. Webpages can contain text, images, links, and multimedia content. In this experiment, a webpage is created to represent a college website, with navigation links that allow users to move between different sections easily.

XAMPP

XAMPP is a free, open-source local web server package that includes Apache, MySQL, PHP, and Perl. It allows web pages to be developed and tested locally on a computer without requiring an internet connection. In this experiment, XAMPP is used to host and execute HTML files using the localhost server.

STEPS:

- **Create the HTML structure** (Start with a basic HTML file structure)
- **Create sections for each content area**
- **Style your web page** (Use CSS to style your web page as desired.)
- **Add more details** (Enhance the sections with more details, images, and other content as needed.)
- **Link the sections in the navigation** (Update the `<a>` tags in the navigation to link to the corresponding sections using the id attribute)
- **Test your web page** (Test your web page to ensure that all links and content are working as expected.)

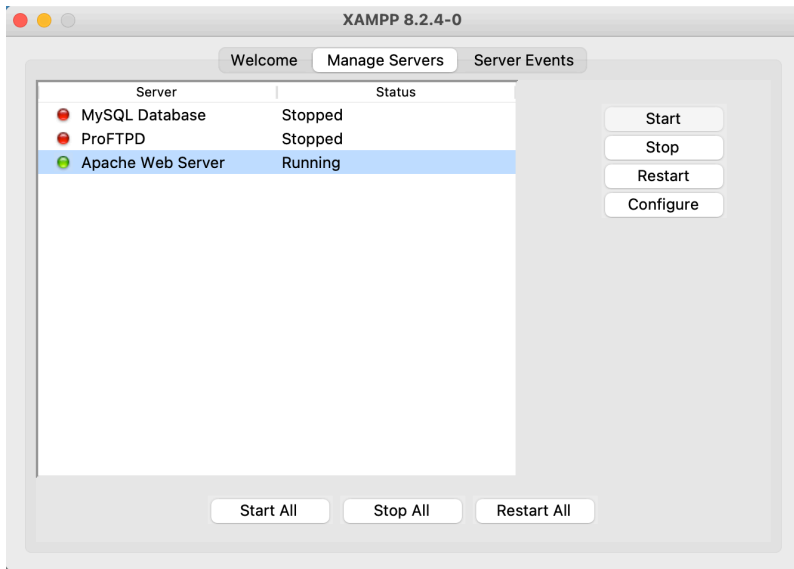
INSTALLATION OF XAMPP

1. Open a web browser and visit the official Apache Friends website. Select XAMPP for your OS and download the latest 64-bit version of the software.

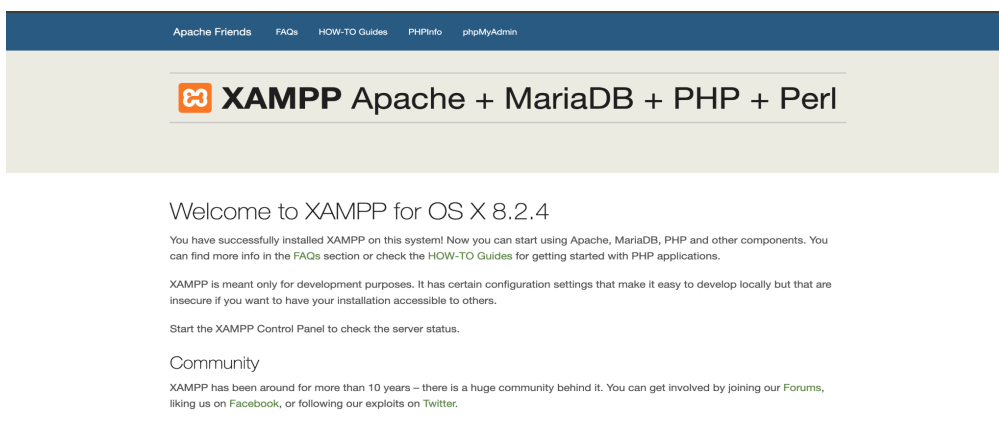


2. Open the downloaded file and move the XAMPP application to the Applications folder.

3. Open the XAMPP Control Panel from the Applications folder. Click on Manage Servers and start the Apache Web Server. Ensure that the server status changes to running.



4. Verify Installation. Open a web browser and type `http://localhost` in the address bar. The appearance of the XAMPP welcome page confirms successful installation.



FOLDER STRUCTURE

Navigate to the `htdocs` directory inside the XAMPP installation folder. Create a new project folder and save all the HTML files related to the experiment in this folder.

CODE :

```
<!DOCTYPE html>
<html>

<head>
  <title>ABC College</title>

  <!-- Internal CSS -->
  <style>
    body {
      font-family: Arial, sans-serif;
      background-color: #f2f2f2;
      margin: 0;
    }

    header {
      background-color: #003366;
      color: white;
      text-align: center;
      padding: 20px;
    }

    nav {
      background-color: #0055aa;
      padding: 10px;
      text-align: center;
    }

    nav a {
      color: white;
      text-decoration: none;
      margin: 15px;
      font-weight: bold;
    }

    nav a:hover {
      text-decoration: underline;
    }

    section {
      background-color: white;
      margin: 20px;
      padding: 20px;
      border-radius: 5px;
    }

    h2 {
      color: #003366;
    }

    footer {
      background-color: #003366;
      color: white;
      text-align: center;
      padding: 10px;
    }
  </style>
</head>

<body>

  <!-- Header -->
  <header>
    <h1>ABC College of Engineering</h1>
    <p>Excellence in Education</p>
  </header>
```

```
<!-- Navigation -->
<nav>
  <a href="#courses">Courses</a>
  <a href="#departments">Departments</a>
  <a href="#faculty">Faculty</a>
  <a href="#library">Library</a>
</nav>

<!-- Courses Section -->
<section id="courses">
  <h2>Courses Offered</h2>
  <ul>
    <li>Bachelor of Technology (B.Tech)</li>
    <li>Master of Technology (M.Tech)</li>
    <li>Bachelor of Science (B.Sc)</li>
    <li>Master of Science (M.Sc)</li>
    <li>MBA</li>
  </ul>
</section>

<!-- Departments Section -->
<section id="departments">
  <h2>Departments</h2>
  <ol>
    <li>Computer Science Engineering</li>
    <li>Information Technology</li>
    <li>Mechanical Engineering</li>
    <li>Electrical Engineering</li>
    <li>Electronics & Communication</li>
  </ol>
</section>

<!-- Faculty Section -->
<section id="faculty">
  <h2>Faculty Members</h2>
  <ul>
    <li>Dr. A. Sharma – Computer Science</li>
    <li>Prof. B. Verma – Mechanical Engineering</li>
    <li>Dr. C. Rao – Electrical Engineering</li>
    <li>Ms. D. Singh – Mathematics</li>
  </ul>
</section>

<!-- Library Section -->
<section id="library">
  <h2>Library</h2>
  <p>
    Our college library has books, journals,
    e-resources, and digital learning facilities.
  </p>

  <ul>
    <li>Reference Books</li>
    <li>National & International Journals</li>
    <li>Digital Library</li>
    <li>Reading Hall</li>
  </ul>
</section>

<!-- Footer -->
<footer>
  © 2026 ABC College. All Rights Reserved.
</footer>

</body>

</html>
```

SNAPSHOTS OF WEBPAGE

ABC College of Engineering

Excellence in Education

[Courses](#)[Departments](#)[Faculty](#)[Library](#)

Courses Offered

- Bachelor of Technology (B.Tech)
- Master of Technology (M.Tech)
- Bachelor of Science (B.Sc)
- Master of Science (M.Sc)
- MBA

Departments

1. Computer Science Engineering
2. Information Technology
3. Mechanical Engineering
4. Electrical Engineering
5. Electronics & Communication

Faculty Members

- Dr. A. Sharma – Computer Science
- Prof. B. Verma – Mechanical Engineering
- Dr. C. Rao – Electrical Engineering
- Ms. D. Singh – Mathematics

Library

Our college library has books, journals, e-resources, and digital learning facilities.

- Reference Books
- National & International Journals
- Digital Library
- Reading Hall

© 2026 ABC College. All Rights Reserved.

EXPERIMENT - 2

AIM - Write html code to develop a webpage having two frames that divide the webpage into two equal rows and then divide the row into equal columns and fill each frame with a different background color.

OBJECTIVE:

To create an HTML webpage using frames that divides the page into two equal rows and further splits each row into equal columns, with each frame having a unique background color.

INTRODUCTION:

This experiment helps students understand how to use the **<frameset>** and **<frame>** tags to design multi-frame web pages. By applying background colors, students will enhance their skills in creating visually distinct and organized layouts.

The **<frameset>** tag in HTML is used to divide a webpage into multiple sections called frames by specifying rows or columns. The **<frame>** tag is used inside a **<frameset>** to load separate HTML pages into each frame. This allows different content to be displayed in different parts of the same browser window. Frames are mainly used to create structured layouts, though they are now outdated and replaced by modern CSS layouts.

STEPS:

- Create the HTML file (Start with a basic HTML file structure)
- Create the frame content (Create two separate HTML files, frame1.html and frame2.html, for the content of each frame)
- View the webpage (Open the main HTML file in a web browser to view the webpage with two frames)
- Adjust frame sizes (Modify the rows attribute in the **<frameset>** tag to adjust the size of the frames. For equal columns, use cols attribute instead.)
- Modify frame content (Update the content in frame1.html and frame2.html to display the desired content in each frame.)

CODE :

exp2.html

```
<!DOCTYPE html>
<html>
<head>
  <title>Two Rows and Columns</title>
</head>

<frameset rows="50%,50%">

  <frameset cols="50%,50%">
    <frame src="frame1.html">
    <frame src="frame2.html">
  </frameset>

  <frameset cols="50%,50%">
    <frame src="frame3.html">
    <frame src="frame4.html">
  </frameset>

</frameset>

</html>
```

frame1.html

```
<!DOCTYPE html>
<html>
<body style="background-color: lightcoral;">
<h2>Frame 1</h2>
</body>
</html>
```

frame2.html

```
<!DOCTYPE html>
<html>
<body style="background-color: lightblue;">
<h2>Frame 2</h2>
</body>
</html>
```

frame3.html

```
<!DOCTYPE html>
<html>
<body style="background-color: lightgreen;">
<h2>Frame 3</h2>
</body>
</html>
```

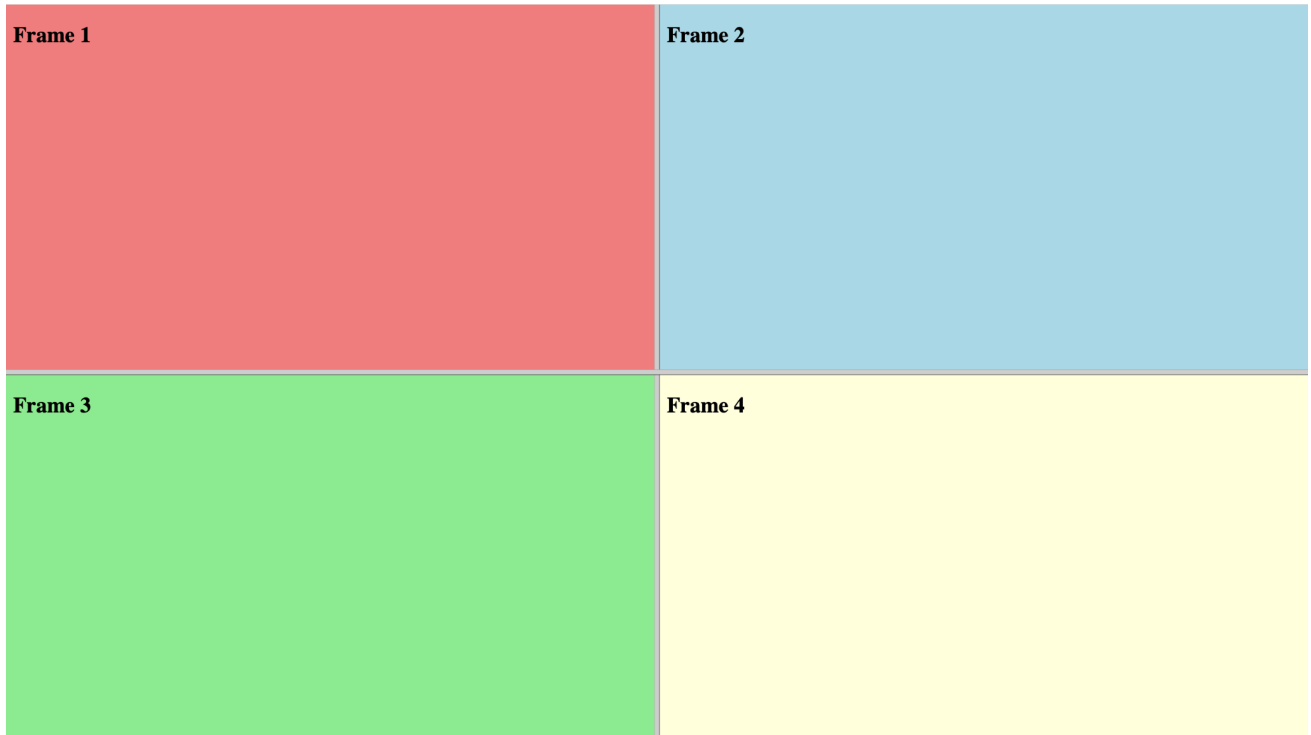
frame4.html

```
<!DOCTYPE html>
```



```
<html>
<body style="background-color: lightyellow;">
<h2>Frame 4</h2>
</body>
</html>
```

OUTPUT :



EXPERIMENT - 3

AIM - Design a web page of your home town with an attractive background color, text color, an Image, font etc. (use internal CSS).

OBJECTIVE:

To design a webpage showcasing your hometown using internal CSS for styling with an attractive background color, text color, fonts, and images.

INTRODUCTION:

This experiment aims to teach students how to use internal CSS to create visually appealing web pages. Students will learn to enhance the aesthetics of a webpage by customizing colors, fonts, and images while structuring the content effectively.

Internal CSS is used to apply styles inside a single HTML page by writing CSS rules within the `<style>` tag in the `<head>` section. It helps control colors, fonts, spacing, and layout without using a separate CSS file, but the styles are limited to that page only.

STEPS:

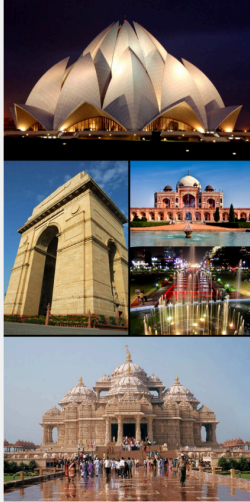
- Create the HTML file (Start with a basic HTML file structure)
- Add content and images (Replace the placeholder text and image (hometown.jpg) with the actual content and image of your hometown.)
- Adjust styles as needed (Modify the styles (background color, text color, font, etc.) to match the design you want for your webpage.)
- View the webpage (Open the HTML file in a web browser to view the webpage with your hometown information)
- Further customization (You can further customize the page by adding more content, styling elements, or using CSS animations and transitions for a more dynamic look.)

CODE:

```
<!DOCTYPE html>
<html>
<head>
  <style>
    body {
      background-color: #f0f0f0;
      font-family: 'Times New Roman', Times, serif;
    }
    .container {
      width: 80%;
      margin: auto;
      text-align: center;
    }
    img {
      width: 300px;
      height: 600px;
    }
    .content {
      display: flex;
      column-gap : 50px;
    }
    p{
      font-size: 28px;
      color: red;
    }
  </style>
</head>
<body>
  <div class="container">
    <h1>DELHI</h1>
    <div class="content">
      
      <p>
        Delhi, officially the National Capital Territory (NCT) of Delhi , is a city and a union territory of India containing New Delhi, the capital of India. Delhi, the capital of India, is adorned with iconic monuments such as the Red Fort, Qutub Minar, and India Gate, reflecting its rich historical legacy. The city pulsates with a diverse culture, showcasing a harmonious blend of traditional customs and modern influences. Delhi's lifestyle is a vibrant tapestry of bustling markets, culinary delights, and a dynamic arts scene, encapsulating the spirit of both ancient heritage and contemporary dynamism.
      </p>
    </div>
  </div>
</body>
</html>
```

OUTPUT :

DELHI



Delhi, officially the National Capital Territory (NCT) of Delhi, is a city and a union territory of India containing New Delhi, the capital of India. Delhi, the capital of India, is adorned with iconic monuments such as the Red Fort, Qutub Minar, and India Gate, reflecting its rich historical legacy. The city pulsates with a diverse culture, showcasing a harmonious blend of traditional customs and modern influences.

Delhi's lifestyle is a vibrant tapestry of bustling markets, culinary delights, and a dynamic arts scene, encapsulating the spirit of both ancient heritage and contemporary dynamism.

EXPERIMENT - 4

AIM - Use External, Internal, and Inline CSS to format the college web page that you created.

OBJECTIVE:

To format a college webpage using External, Internal, and Inline CSS, showcasing the effective use of different CSS techniques for styling and layout.

INTRODUCTION:

This experiment helps students understand and apply the three types of CSS External, Internal, and Inline—to design and style a college webpage. It emphasizes the flexibility and advantages of each CSS method in creating well-structured and visually appealing web pages.

There are three types of CSS used for styling web pages: **Inline CSS, Internal CSS, and External CSS**. Inline CSS is applied directly inside an HTML element using the style attribute and is used for specific elements. Internal CSS is written inside the <style> tag in the <head> section and is used to style a single webpage. External CSS is written in a separate .css file and linked to the HTML page, allowing the same styles to be reused across multiple webpages and making maintenance easier.

STEPS:

- **External CSS** (Create an external CSS file named style.css to define styles for the entire website)
- **Internal CSS (in the <head> section of your HTML file)** (Add internal CSS to customize specific elements or sections of the page)
- **Link external CSS file** (Add a link to your external CSS file in the <head> section of your HTML file)
- **Use classes and IDs for styling** (Add classes or IDs to HTML elements to apply styles defined in the external or internal CSS)
- **Inline CSS (within specific HTML elements)** (Use inline CSS to style individual elements directly in the HTML file.)

CODE :

exp4.html

```
<!DOCTYPE html>
<html>

<head>
  <title>ABC College</title>

  <!-- External CSS File -->
  <link rel="stylesheet" href="style.css">

  <!-- Internal CSS -->
  <style>
    h1 {
      text-align: center;
      color: white;
    }

    h2 {
      color: #003366;
    }

    .box {
      background-color: white;
      margin: 20px;
      padding: 15px;
      border-radius: 5px;
    }
  </style>
</head>

<body style="font-family: Arial; background-color: #f2f2f2;">

  <!-- Inline CSS -->
  <header style="background-color:#003366; color:white; padding:20px; text-align:center;">
    <h1>ABC College of Engineering</h1>
    <p>Excellence in Education</p>
  </header>

  <!-- Navigation -->
  <nav>
    <a href="#courses">Courses</a>
    <a href="#departments">Departments</a>
    <a href="#faculty">Faculty</a>
    <a href="#library">Library</a>
  </nav>

  <!-- Courses -->
  <section id="courses" class="box">
    <h2>Courses Offered</h2>
    <ul>
      <li>B.Tech</li>
      <li>M.Tech</li>
      <li>B.Sc</li>
      <li>MBA</li>
    </ul>
  </section>

  <!-- Departments -->
```

```

<section id="departments" class="box">
  <h2>Departments</h2>
  <ol>
    <li>Computer Science</li>
    <li>Mechanical</li>
    <li>Electrical</li>
    <li>Electronics</li>
  </ol>
</section>

<!-- Faculty -->
<section id="faculty" class="box">
  <h2>Faculty Members</h2>
  <ul>
    <li>Dr. Sharma</li>
    <li>Prof. Verma</li>
    <li>Ms. Singh</li>
  </ul>
</section>

<!-- Library -->
<section id="library" class="box">
  <h2>Library</h2>
  <p>
    Our library contains books, journals,
    and digital learning resources.
  </p>
</section>

<!-- Footer -->
<footer style="background-color:#003366; color:white; text-align:center; padding:10px;">
  © 2026 ABC College
</footer>

</body>

</html>

```

style.css

```

nav {
  background-color: #0055aa;
  padding: 10px;
  text-align: center;
}

nav a {
  color: white;
  text-decoration: none;
  margin: 15px;
  font-weight: bold;
}

nav a:hover {
  text-decoration: underline;
}

```

OUTPUT:

ABC College of Engineering

Excellence in Education

Courses

Departments

Faculty

Library

Courses Offered

- B.Tech
- M.Tech
- B.Sc
- MBA

Departments

- Computer Science
- Mechanical
- Electrical
- Electronics

Faculty Members

- Dr. Sharma
- Prof. Verma
- Ms. Singh

Library

Our library contains books, journals, and digital learning resources.

© 2026 ABC College

EXPERIMENT - 5

AIM - Create HTML Page with JavaScript which takes an Integer number as input and tells whether the number is ODD or EVEN.

OBJECTIVE:

To create an HTML page with JavaScript that accepts an integer as input and determines whether the number is odd or even.

INTRODUCTION:

This experiment focuses on integrating JavaScript with HTML to build interactive web pages. Students will learn to take user input, process it using conditional logic in JavaScript, and display the result dynamically on the webpage.

JavaScript is integrated into an HTML page using the `<script>` tag, which allows adding interactivity and dynamic behavior to the webpage. The script can be written inside the HTML file or linked as an external JavaScript file.

STEPS:

- **Create the HTML file** (Start with a basic HTML file structure)
- **Add CSS for styling** (Add CSS to style the elements for better presentation)
- **Add JavaScript code** (Add JavaScript code to check if the input number is odd or even)
- **Test the webpage** (1. Open the HTML file in a web browser. 2. Enter an integer number in the input field and click the "Check" button to see if the number is odd or even.)

CODE :

```
<!DOCTYPE html>
<html>
<head>
  <title>Odd or Even</title>
</head>
<body>
  <h1>Odd or Even</h1>
  <input type="number" id="number" placeholder="Enter a number">
  <button onclick="checkNumber()">Check</button>
  <p id="result"></p>

  <script>
    function checkNumber() {
      let num = document.getElementById('number').value;
      if (num % 2 == 0) {
        document.getElementById('result').innerHTML = num + " is an Even number";
      } else {
        document.getElementById('result').innerHTML = num + " is an Odd number";
      }
    }
  </script>
</body>
</html>
```

OUTPUT:

Odd or Even

5 is an Odd number

EXPERIMENT - 6

AIM - Create HTML Page that contains form with fields Name, Email, Mobile No, Gender , Favourite Colour and a button now write a JavaScript code to combine and display the information in textbox when the button is clicked and implement validation.

OBJECTIVE:

To create an HTML form with fields for Name, Email, Mobile No, Gender, Favorite Color, and a button, using JavaScript to display the combined input in a textbox and implement validation.

INTRODUCTION:

This experiment teaches students to create dynamic web forms and validate user input using JavaScript. They will also learn to display the combined information in a textbox, improving their form handling and interactivity skills.

JavaScript is integrated into an HTML page using the `<script>` tag, which allows adding interactivity and dynamic behavior to the webpage. The script can be written inside the HTML file or linked as an external JavaScript file.

STEPS:

- **Create an HTML file** with the following structure (Name, Email, Mobile No, Gender , Favorite Color)
- **Add JavaScript code** (Add JavaScript code to combine and display the information in the textbox)
- **Implement validation** (1. Add required attribute to the input fields to make them required. 2. You can also add additional validation logic in the combineInfo() function as needed.)
- **Test the webpage** (1. Open the HTML file in a web browser. 2. Fill in the form fields and click the "Submit" button to see the combined information displayed in the textbox.)

CODE :

```
<!DOCTYPE html>
<html>
<head>
  <title>Form</title>
  <style>
    form{
      display: flex;
      flex-direction: column;
      width: 200px;
      gap: 10px;
    }

  </style>
</head>
<body>
  <form id="myForm">
    <label for="name">Name:</label>
    <input type="text" id="name" name="name" required>
    <label for="email">Email:</label>
    <input type="email" id="email" name="email" required>
    <label for="mobile">Mobile No:</label>
    <input type="tel" id="mobile" name="mobile" pattern="[0-9]{10}" required>
    <label for="gender">Gender:</label>
    <select id="gender" name="gender" required>
      <option value="">Select...</option>
      <option value="male">Male</option>
      <option value="female">Female</option>
    </select>
    <label for="color">Favourite Colour:</label>
    <input type="color" id="color" name="color" required>
    <input type="button" value="Submit" onclick="displayInfo()">
  </form>
  <p id="info"></p>

  <script>
    function displayInfo() {
      var form = document.getElementById('myForm');
      if (!form.checkValidity()) {
        alert('Please fill out all fields correctly.');
```

Favourite Colour: ' + color;

```
        document.getElementById('info').textContent = info;
      }
    }
  </script>
</body>
</html>
```

OUTPUT:

Name:

Email:

Mobile No:

Gender:

Favourite Colour:



Name: yash, Email: abc@gmail.com, Mobile: 7894561230, Gender: male, Favourite Colour: #dc1818

EXPERIMENT - 7

AIM - Create XML file to store student information like Enrolment Number, Name Mobile Number , Email Id.

OBJECTIVE:

To create an XML file that stores student information such as Enrolment Number, Name, Mobile Number, and Email ID.

INTRODUCTION:

This experiment helps students understand how to organize and store data in XML format. Students will gain hands-on experience in structuring information using XML tags and attributes.

XML (eXtensible Markup Language) is a markup language used to store, organize, and transport data in a structured and readable format. It is both human-readable and machine-readable, making it widely used for data exchange between systems.

STEPS:

- **Create the XML file** (Create a new file with a .xml extension (e.g., students.xml).
- **Define the XML structure** (Use the structure to define the XML file with student information)
- **Save the XML file** (Save the file with the .xml extension (e.g., students.xml).
- **Add more student entries** (To add more student entries, copy the <student> block and update the information accordingly.)
- **Test the XML file** (Open the XML file in a web browser or an XML viewer to ensure that the structure and data are displayed correctly.)

CODE :

exp7.xml

```
<?xml-stylesheet type="text/css" href="styles.css"?>

<!DOCTYPE HTML>
<html>

<head>
  <h1> STUDENTS DESCRIPTION </h1>
</head>

<students>

  <student>
    <enrollmentno>Enrollment No. : 4AB16CS001</enrollmentno>
    <name>Name : Yash</name>
    <mobile>Mobile : 9876541230</mobile>
    <e-mail>E-Mail : abc@gmail.com</e-mail>
  </student>

  <student>
    <enrollmentno>Enrollment No. : 4AB17CS002</enrollmentno>
    <name>Name : Hari</name>
    <mobile>Mobile : 9756522330</mobile>
    <e-mail>E-Mail : xyz@gmail.com</e-mail>
  </student>

  <student>
    <enrollmentno>Enrollment No. : 4AB18CS003</enrollmentno>
    <name>Name : Raj</name>
    <mobile>Mobile : 9581184193</mobile>
    <e-mail>E-Mail : pqr@gmail.com</e-mail>
  </student>

</students>

</html>
```

styles.css

```
students {
  display: block;
  margin: 20px;
  font-family: Times New Roman;
}

heading {
  display: block;
  font-size: 20px;
  font-weight: bold;
  margin-bottom: 20px;
}

student {
  display: block;
  margin-bottom: 25px;
}

enrollmentno,
name,
mobile,
```

```
email {  
  display: block;  
  font-size: 18px;  
}
```

OUTPUT:

STUDENTS DESCRIPTION

Enrollment No. : 4AB16CS001
Name : Yash
Mobile : 9876541230
E-Mail : abc@gmail.com

Enrollment No. : 4AB17CS002
Name : Hari
Mobile : 9756522330
E-Mail : xyz@gmail.com

Enrollment No. : 4AB18CS003
Name : Raj
Mobile : 9581184193
E-Mail : pqr@gmail.com

EXPERIMENT - 8

AIM - Write a php script to read data from txt file and display it in html table (the file contains info in format Name: Password: Email)

OBJECTIVE:

To write a PHP script that reads data from a text file containing Name, Password, and Email, and displays the information in an HTML table.

INTRODUCTION:

This experiment focuses on using PHP to read data from a text file and present it in an HTML table. Students will learn how to handle file input and organize the content for display within a web page.

PHP (Hypertext Preprocessor) is a server-side scripting language mainly used for web development to create dynamic and interactive web pages.

It is embedded within HTML and executed on the server, generating content that is sent to the browser.

PHP supports databases, sessions, and form handling, making it widely used for backend development.

STEPS:

- **Create a text file (data.txt)** (Create a text file named data.txt and add some sample data in the format Name:Password:Email on each line)
- **Write the PHP script** (Create a PHP script named xyz.php to read the data from the text file and display it in an HTML table)
- **Save the PHP script** (Save the xyz.php file in the same directory as the data.txt file.)
- **Access the PHP script in a web browser** (Open a web browser and navigate to <http://localhost/xyz.php> to see the data from the data.txt file displayed in an HTML table.)

CODE :

exp8.php

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>User Data</title>
</head>
<body>

<?php
$file = fopen("file1.txt", "r") or die("Unable to open file!");

echo '<table border="1">
    <tr>
        <th>Name</th>
        <th>Password</th>
        <th>Email</th>
    </tr>';

while (!feof($file)) {
    $line = trim(fgets($file));

    if (!empty($line)) {
        $data = explode(":", $line);

        if (count($data) == 3) {
            echo "<tr>
                <td>{$data[0]}</td>
                <td>{$data[1]}</td>
                <td>{$data[2]}</td>
            </tr>";
        }
    }
}

echo '</table>';
fclose($file);
?>

</body>
</html>
```

file1.txt

Yash:pass123:yash@gmail.com
Riya:riya456:riya@gmail.com
Aman:aman789:aman@gmail.com
Neha:neha321:neha@gmail.com

OUTPUT:

Name	Password	Email
Yash	pass123	yash@gmail.com
Riya	riya456	riya@gmail.com
Aman	aman789	aman@gmail.com
Neha	neha321	neha@gmail.com

EXPERIMENT - 9

AIM - Write a PHP Script for login authentication. Design an html form which takes username and password from user and validate against stored username and password in file.

OBJECTIVE:

To write a PHP script for login authentication that validates the username and password entered by the user against stored credentials in a file.

INTRODUCTION:

This experiment involves designing an HTML login form and implementing PHP to securely validate user input based on stored data.

Login authentication is a fundamental concept in web security that ensures only authorized users can access a system. It typically involves verifying user credentials against stored data using server-side scripting.

PHP provides built-in functions to handle form data, sessions, and file handling for authentication.

STEPS:

- **Create a text file** (Create a text file named users.txt and add some sample username and password combinations in the format username:password on each line)
- **Write the PHP script** (Create a PHP script named login.php to handle the login authentication)
- **Save the PHP script** (Save the login.php file in the same directory as the users.txt file)
- **Access the login form in a web browser** (1. Open a web browser and navigate to <http://localhost/login.php> to see the login form. 2. Enter a valid username and password from the users.txt file and click the "Login" button to see the login result.)

CODE :

exp9.php

```
<!DOCTYPE html>
<html>
<body>
<h2>Login Form</h2>
<form method="post" action="<?php echo $_SERVER['PHP_SELF'];?>">
  Username:<br>
  <input type="text" name="username"><br>
  Password:<br>
  <input type="password" name="password"><br><br>
  <input type="submit" value="Submit"> <br><br>
</form>

<?php
if ($_SERVER["REQUEST_METHOD"] == "POST") {
    $username = $_POST["username"];
    $password = $_POST["password"];

    $file = fopen("credentials.txt", "r");
    $correctCredentials = false;

    while(!feof($file)) {
        $line = fgets($file);
        list($fileUsername, $filePassword) = explode(",", $line);
        if(trim($fileUsername) == $username && trim($filePassword) == $password) {
            $correctCredentials = true;
            break;
        }
    }

    fclose($file);
    if($correctCredentials) {
        echo "Login successful!";
    } else {
        echo "Invalid username or password!";
    }
}
?>

</body>
</html>
```

credentials.txt

```
admin,1234
user,test123
abc,pass
```

OUTPUT:

Login Form

Username:

Password:

Login Form

Username:

Password:

Login successful!

EXPERIMENT - 10

AIM -Write PHP Script for storing and retrieving user information from MySql table.

OBJECTIVE:

To write a PHP script that stores user information in a MySQL table and retrieves it for display.

INTRODUCTION:

This experiment helps students learn how to connect PHP with MySQL, allowing them to store and retrieve data using SQL queries.

STEPS:

- a. Design A HTML page which takes Name, Address, Email and Mobile No. From user **(register.php)**
- b. Store this data in Mysql database / text file.
- c. Next page display all user in html table using PHP **(display.php)**
 - **Create a MySQL database and table** (Create a MySQL database (e.g., user_info) and a table (e.g., users) with columns for Name, Address, Email, and Mobile No.)
 - **Design the HTML form (register.php)** (Create an HTML form to collect user information)
 - **Store user information in the MySQL database (register.php)** (Create a PHP script to process the form data and insert it into the MySQL database)
 - **Display all users in an HTML table (display.php)** (Create a PHP script to retrieve user information from the MySQL database and display it in an HTML table)
 - **Access the HTML form and display page in a web browser** (Access register.php to register new users and display.php to display all users in an HTML table.)

CODE :

register1.php

```
<!-- register.php -->
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>User Registration</title>
</head>
<body>
```

```

<h2>User Registration</h2>
<form method="post" action="<?php echo $_SERVER['PHP_SELF'];?>">
    <label>Name:</label>
    <input type="text" name="name" required><br><br>

    <label>Address:</label>
    <input type="text" name="address" required><br><br>

    <label>Email:</label>
    <input type="email" name="email" required><br><br>

    <label>Mobile No.:</label>
    <input type="text" name="mobile" required><br><br>

    <input type="submit" value="Register"><br>
</form>
<?php
    if ($_SERVER["REQUEST_METHOD"] == "POST") {
        $conn = new mysqli('localhost','root','', 'test1');
        if ($conn->connect_error) {
            die("Connection failed: " . $conn->connect_error);
        }

        $name = $_POST['name'];
        $address = $_POST['address'];
        $email = $_POST['email'];
        $mobile = $_POST['mobile'];
        $sql = "INSERT INTO users (name, address, email, mobile) VALUES ('$name', '$address', '$email', '$mobile')";
        if ($conn->query($sql) === TRUE) {
            echo 'User registered successfully';
        } else {
            echo "Error: " . $sql . "<br>" . $conn->error;
        }

        $conn->close();
    }
?>

</body>
</html>

```

display1.php

```

<?php
$conn = new mysqli('localhost', 'root', '', 'test1');
if ($conn->connect_error) {
    die("Connection failed: " . $conn->connect_error);
}

```



```

$result = $conn->query('SELECT * FROM users');
echo '<table border="1">
    <tr>
        <th>Name</th>
        <th>Address</th>
        <th>Email</th>
        <th>Mobile No</th>
    </tr>';

while ($row = $result->fetch_assoc()) {
    echo '<tr>
        <td>' . $row['name'] . '</td>
        <td>' . $row['address'] . '</td>
        <td>' . $row['email'] . '</td>
        <td>' . $row['mobile'] . '</td>
    </tr>';
}

echo '</table>';
$conn->close();
?>

```

In test1 database in phpmyadmin, create users table via sql tab

```

CREATE TABLE users (
    name VARCHAR(100),
    address VARCHAR(200),
    email VARCHAR(100),
    mobile VARCHAR(20)
);

```

then run register1 page and display page seperately

OUTPUT:

User Registration

Name:

Address:

Email:

Mobile No.:

Register

User Registration

Name:

Address:

Email:

Mobile No.:

Register

User registered successfully

```
MariaDB [test]> select * from users;
+----+-----+-----+-----+-----+
| id | name | address | email | mobile |
+----+-----+-----+-----+-----+
| 5  | yash | 123 Lane | abc@gmail.com | 9789456213 |
| 6  | abc  | xyz lane | 123@gmail.com | 7894561230 |
+----+-----+-----+-----+-----+
2 rows in set (0.031 sec)
```

Name	Address	Email	Mobile No
yash	123 Lane	abc@gmail.com	9789456213
abc	xyz lane	123@gmail.com	7894561230

EXPERIMENT - 1 (BEYOND SYLLABUS)

AIM -Design a simple calculator using JavaScript.

OBJECTIVE:

Design a simple calculator using JavaScript.

INTRODUCTION:

This experiment teaches students how to use JavaScript to create an interactive calculator. Students will learn how to handle user input and implement arithmetic functions to perform calculations dynamically on a webpage.

STEPS:

- **Create the HTML Structure for the Calculator** (basic structure of the calculator with buttons for digits, operators, and a display area.)
- **Add CSS to Style the Calculator** (layout, colors, button styles, and text alignment.)
- **Add JavaScript to Handle Calculator Functionality** (user input, perform calculations, and update the display.)

CODE :

```
<!DOCTYPE html>
<html>
<head>
  <title>Calculator</title>
  <style>
    body {
      text-align: center;
      font-family: Arial;
    }

    h1 {
      margin-top: 20px;
    }

    .calculator {
      width: 220px;
      margin: auto;
    }

    #display {
      width: 100%;
      height: 50px;
      font-size: 20px;
      text-align: right;
      margin-bottom: 10px;
    }

    .buttons {
```

```

        display: grid;
        grid-template-columns: repeat(4, 1fr);
        gap: 5px;
    }

    button {
        height: 50px;
        font-size: 16px;
        cursor: pointer;
    }
</style>
</head>
<body>

<h1>Calculator</h1>

<div class="calculator">
    <input type="text" id="display" disabled>

    <div class="buttons">
        <button onclick="clearDisplay()">C</button>
        <button onclick="append('*2')">x2</button>
        <button onclick="append('%')">%</button>
        <button onclick="append('/')">/</button>

        <button onclick="append('7')">7</button>
        <button onclick="append('8')">8</button>
        <button onclick="append('9')">9</button>
        <button onclick="append('*')">X</button>

        <button onclick="append('4')">4</button>
        <button onclick="append('5')">5</button>
        <button onclick="append('6')">6</button>
        <button onclick="append('-')">-</button>

        <button onclick="append('1')">1</button>
        <button onclick="append('2')">2</button>
        <button onclick="append('3')">3</button>
        <button onclick="append('+')">+</button>

        <button onclick="append('0')">0</button>
        <button onclick="append('00')">00</button>
        <button onclick="append('.')">.</button>
        <button onclick="calculate()">=</button>
    </div>
</div>

<script>
function append(value) {
    document.getElementById("display").value += value;
}

function clearDisplay() {
    document.getElementById("display").value = "";
}

function calculate() {
    try {
        let result = eval(document.getElementById("display").value);
    }

```

```
        document.getElementById("display").value = result;
    } catch {
        document.getElementById("display").value = "Error";
    }
}
</script>

</body>
</html>
```

OUTPUT:

Calculator

13+12			
C	x2	%	/
7	8	9	X
4	5	6	-
1	2	3	+
0	00	.	=

EXPERIMENT - 2 (BEYOND SYLLABUS)

AIM -Design a website that displays a quiz for students and calculates the score using JavaScript after the user submits.

OBJECTIVE:

To design a website that presents a quiz to students and calculates their score using JavaScript after submission.

INTRODUCTION:

To design a website that presents a quiz to students and calculates their score using JavaScript after submission.

STEPS:

- **Create the HTML Structure for the Quiz** (quiz questions and answer options.)
- **Add CSS for Styling** (quiz visually appealing)
- **Write JavaScript to Calculate the Score** (logic that will calculate the score based on the user's answers when the "Submit" button is clicked.)

CODE :

```
<!DOCTYPE html>
<html>
<head>
  <title>Quiz</title>
  <style>
    body {
      font-family: Arial;
      text-align: center;
    }
    .quiz-box {
      width: 400px;
      margin: auto;
      text-align: left;
    }
    button {
      margin-top: 10px;
      padding: 8px;
    }
  </style>
</head>
<body>

<h2>QUIZ</h2>

<div class="quiz-box">
  <form id="quizForm">

    <p><b>Q1 - A constructor can be virtual?</b></p>
```

```

<input type="radio" name="q1" value="True"> True<br>
<input type="radio" name="q1" value="False"> False<br>

<p><b>Q2 - Which data type can be used to hold a wide character in C++?</b></p>
<input type="radio" name="q2" value="unsigned char"> unsigned char<br>
<input type="radio" name="q2" value="int"> int<br>
<input type="radio" name="q2" value="wchar_t"> wchar_t<br>
<input type="radio" name="q2" value="None"> None of the above<br>

<button type="button" onclick="calculateScore()">Submit</button>
</form>

<h3 id="result"></h3>
</div>

<script>
function calculateScore() {
    let score = 0;

    let q1 = document.querySelector('input[name="q1"]:checked');
    let q2 = document.querySelector('input[name="q2"]:checked');

    if (q1 && q1.value === "False") score++;
    if (q2 && q2.value === "wchar_t") score++;

    document.getElementById("result").innerText = "Your Score: " + score + "/2";
}
</script>

</body>
</html>

```

OUTPUT:

QUIZ

Q1 - A constructor can be virtual?

- ☐ True
☒ False

Q2 - Which data type can be used to hold a wide character in C++?

- ☐ unsigned char
☐ int
☒ wchar_t
☐ None of the above

Submit

Your Score: 2/2